



HYDROGRAPH

RGA-COOH-1 Reactive Graphene Aggregate Carboxylic Acid Type Technical Datasheet

HydroGraph 2101 East St Elmo Rd, Ste 150 Austin, TX 78744 +1- (785) 380-4205	
Product Number: RGA-COOH-1	Product: Reactive Graphene Aggregate Carboxylic Acid Type
Product Description	
A few layer, functionalized, turbostratic graphene in a powdered form of aggregated nanoplatelets from carbon-rich gas explosion synthesis	

Product Information	
Production Method	Controlled Oxidation in Batch Reactor
Raw Material	Fractal Graphene Aggregate
Forms of Materials	PWD – Dry Powder

Characteristic	Test Method	Value
SP2 Bonded Carbon	RAMAN, XPS	Yes (G peak), 100% sp ² (D parameter)
Structural Defects	RAMAN	D/G = 0.66 G width = 39 cm ⁻¹
Number of Layers	X-Ray Diffraction	6-layer average
Z-Axis Dimensions	X-Ray Diffraction	3±0.5nm
Primary Particle Shape	TEM, Light Scattering	Platelets (aggregated)
Lateral Dimensions	TEM	20-50nm
Aspect Ratio	TEM	1:15
Tapped Bulk Density		200-270 mg/mL*
Chemical/Elemental Analysis	XPS	C 96.3%
Oxygen Content %	XPS	2.1%
Impurities %	XPS	1.6%
Functionalization	FTIR	-COOH
Surface Particle Charge	Zeta Potential	-25mV
Graphene Orientation	RAMAN, XRD	Turbostratic
Specific Surface Area (SSA)	BET	130-180 m ² /g
Crystallinity	Electron Diffraction, X-Ray Diffraction	Crystalline

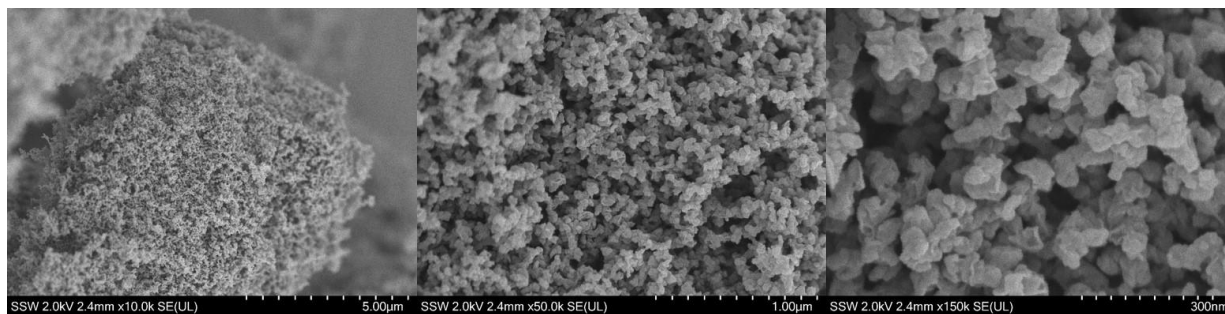
*Density may vary



HYDROGRAPH

Parameters	
Appearance	Black powder
Number of Layers	3-9 layers
Lateral Size	Nano-platelets 20 to 50nm. Aggregates radius of gyration ~150nm
Shape and Form	Fractal aggregate of nano-platelets
Elemental Analysis	Atomic %: 96.3% Carbon, 2.1% Oxygen, 1.6% Hydrogen, No PAHs
Dispersants/Surfactants	None
Concentration	100%
Solid Content	100%
Solvent content	N/A
Substrate Material	N/A
Sheet Resistance	Not applicable
Color	Light absorbing. Black $L^*=3.5$, $a^*=-0.06$, $b^*=-0.68$ (10° observer/D65 Illuminant)
Odor	None
Solubility in Water	Hydrophilic
Electrical Conductivity	Function of powder compression. 100-250 S/m
Thermal Stability	Thermo-gravimetric analysis (TGA) shows: In nitrogen- No volatiles up to 600°C In air- Stable up to 439°C, $T_{max}=638.3^{\circ}\text{C}$ (at $dT/dt=10^{\circ}\text{C}/\text{min}$)

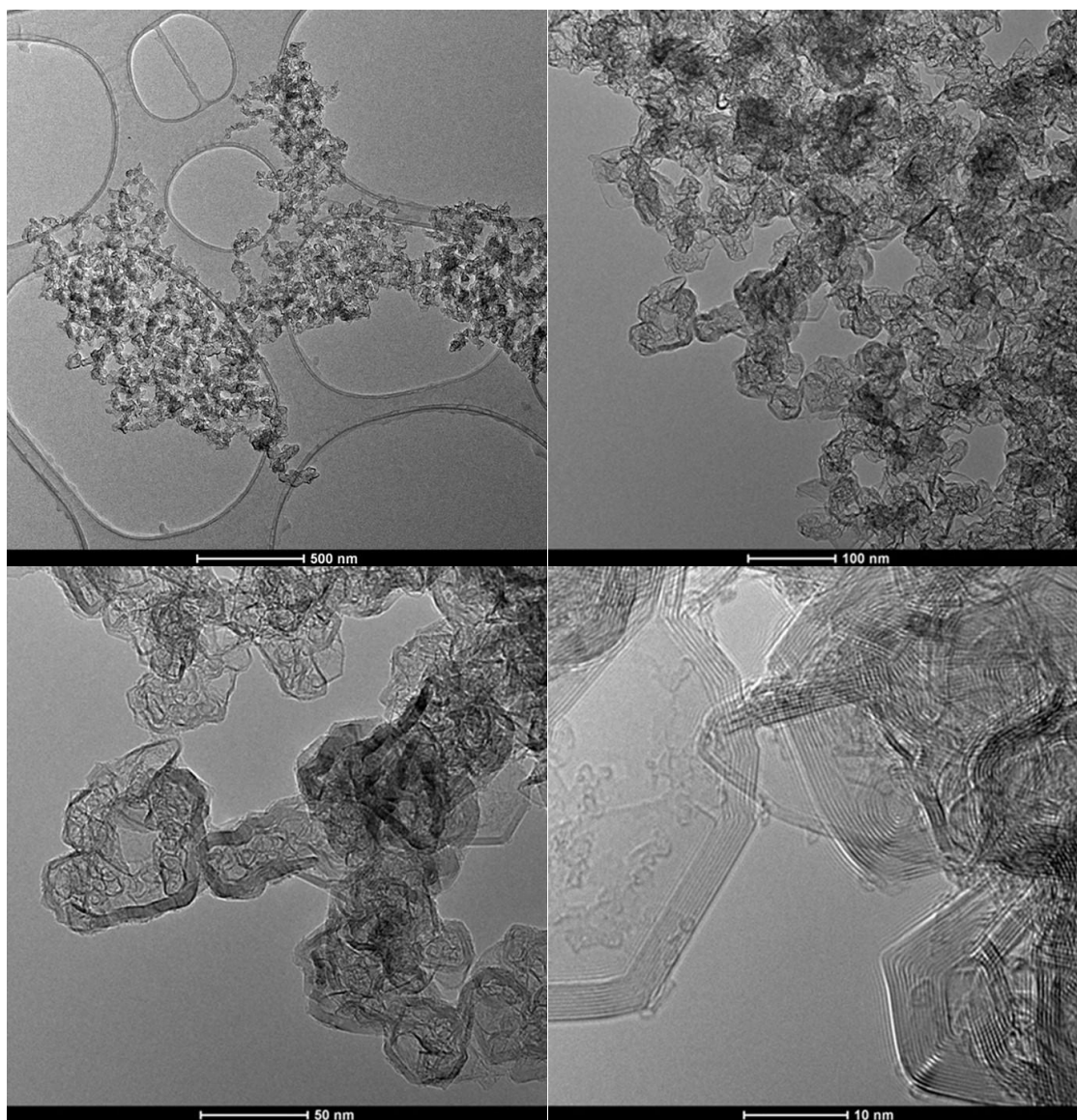
Scanning Electron Micrographs





HYDROGRAPH

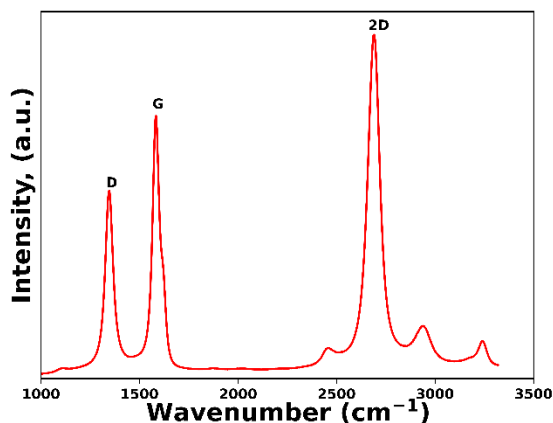
Transmission Electron Micrographs



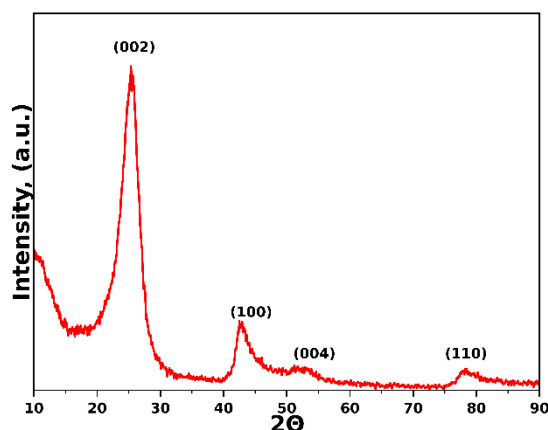


HYDROGRAPH

Raman Spectrum (532 nm)



X-Ray Diffractogram



Notes on Analysis:

- The turbostratic nature is indicated by the asymmetric (100) peak in the X-Ray Diffractogram and the symmetric Lorentzian 2D peak in the Raman Spectrum.
- AFM- The aggregate nature of our fractal graphene is not amenable to AFM analysis.
- Raman- The nanoscale lateral dimensions of our monomer platelets lead to a high fraction of defect edge sites which enhance the intensity of the Raman D bands.

Disclaimer: The technical information and other statements contained in this document are based upon tests Hydrograph believes are reliable, but the accuracy or completeness of such information is not guaranteed. Given the variety of factors that can affect the use and performance of this product, user is solely responsible for evaluating the Hydrograph product and determining whether it is fit for a particular purpose and suitable for user's method of application.